

The Role of Advertising in High-Tech Medical Procedures: Evidence from Robotic Surgeries

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Abstract

Hospital advertising has grown more than five-fold in the past two decades. However, unlike detailing and advertising for prescription drugs, the topic of hospital advertising has been understudied. This research introduces a customer-centric view to this market by investigating the role of advertising in patients' choice of high-tech medical procedures, with a focus on robotic surgery. The authors analyze approximately 140,000 individual patient records and television advertising data from Florida during 2011–2015 to investigate how hospital advertising of robotic surgery affects patients' choice of robotic surgery over more conventional laparoscopic and open surgeries. Using a variation of a designated market area border identification strategy, the authors find that this advertising leads to more robotic surgery choices. The advertising effect is especially strong for Medicaid patients, whose socioeconomic status tends to be lower. While robotic surgery is associated with a short-term health benefit (i.e., reduced length of hospital stay), it does not affect long-term health benefits and comes at a higher cost than other forms of surgery. Thus, understanding the effect of advertising robotic surgery has significant health, cost, and marketing implications for different stakeholders in the health care industry, such as patients, health care providers, surgical robot manufacturers, insurance providers, and policy makers.

Keywords

health care marketing, advertising, causal inference, public policy

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I can't tell you how many patients come in who say, "I want robotic surgery with a laser."... This is a beautiful illustration of how American medicine and the patients tend to become enamored of technology without ever really asking the question, "What are we getting for the technology?"

—A surgeon at Mount Sinai Hospital, New York

The U.S. health care sector has grown steadily in recent years, with total expenditures reaching \$3.6 trillion (17.7% of gross domestic product) in 2018.¹ This growth has been accompanied by a substantial increase in marketing expenditures; health care marketing alone amounted to nearly \$30 billion in 2016 (Schwartz and Woloshin 2019). Health care marketing generally consists of two main activities, depending on the target segment: pharmaceutical marketing to health care

professionals and direct-to-consumer advertising (DTCA) for prescription drugs and health services. The former often includes prescriber detailing (e.g., visits by pharmaceutical representatives to physicians to inform them about new drugs and drug attributes), free samples, and direct physician payments, and this activity accounts for 68% of marketing expenditures. The latter accounts for the remainder of expenditures and involves advertising directed at the public to promote prescription drugs, hospitals, clinics, and other medical services.

Pharmaceutical marketing to health care professionals has garnered considerable academic interest in marketing (e.g., Guo, Sriram, and Manchanda 2020; Iizuka and Jin 2005;

¹ See <https://www.cdc.gov/nchs/fastats/health-expenditures.htm> (accessed on March 1, 2019).

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Kim and KC 2020a; Liu and Gupta 2011; Narayanan, Desiraju, and Chintagunta 2004; Shapiro 2020). One stream of research has examined the intertemporal effects of prescriber detailing (Chan, Narasimhan, and Xie 2013; Ching 2010; Ching and Ishihara 2012; Chintagunta, Jiang, and Jin 2009; Narayanan and Manchanda 2009), such as physicians learning about efficacy, side effects, and other drugs in the drug category. Studies also note informative and persuasive roles of pharmaceutical detailing (Chan, Narasimhan, and Xie 2013; Ching and Ishihara 2012).

Another stream of literature has focused on how consumers are affected by DTCA, which grew substantially between 1997 and 2016 from \$2.1 billion to \$9.6 billion, an increase partly explained by the rise of consumerism in the U.S. health care industry. Driven by health insurance plans with high deductibles and copayments, greater transparency regarding provider quality, and the individual mandate for health care plans, patients have taken more control of decision making in health care products and delivery options (Cordina, Kumar, and Olson 2017; Huckman and Kelley 2013). Studies on DTCA show that consumer advertising affects prescription fills by increasing decisions to get a checkup (Hosken and Wendling 2013) and schedule doctor visits (Iizuka and Jin 2005, 2007; Liu and Gupta 2011). Studies also measure meaningful demand effects in various drug categories (Alpert, Lakdawalla, and Sood 2015; Narayanan, Desiraju, and Chintagunta 2004; Rosenthal et al. 2003; Shapiro 2018; Sinkinson and Starc 2019; Wosinska 2002) and show increased drug adherence to statin drugs due to DTCA (Wosinska 2005). In terms of health insurance advertising, Aizawa and Kim (2018) and Shapiro (2020) examine advertising for Medicare Advantage plans, and Karaca-Mandic et al. (2017) and Aizawa and Kim (2020) investigate the insurance industry's and government's advertising in the case of the Affordable Care Act rollout.

Hospital DTCA soared more than 500% between 1997 and 2016, growing from \$542 million to nearly \$3 billion (Schwartz and Woloshin 2019). Given this remarkable growth, it is surprising that empirical research on hospital marketing practices directed at patients is scarce in marketing, with the exception of Kim and KC (2020b), who investigate the role of general hospital advertising in attracting patients in Massachusetts. Thus, to broaden our understanding of the impact of hospital DTCA, the present study explores the patient choice regarding surgery, which accounts for approximately half of hospital expenditures (Kaye et al. 2020). This investigation is motivated by the fact that hospitals often advertise high-tech procedures such as proton therapy, CyberKnife radiotherapy, and robotic surgery, which are substitutes for traditional procedures. However, our interviews with hospital marketing departments have revealed that measuring the return on such promotion is difficult. With fast-growing hospital advertising expenditures, health care marketers need to understand whether DTCA effectively influences the choice of procedure. To explore this issue, we focus on robotic surgery advertising and empirically quantify the effect of DTCA on the choice of a high-tech procedure over a conventional one.

Surgical procedures can generally be categorized as one of three types: open surgery and two types of minimally invasive surgery, namely laparoscopic surgery and robotic surgery. Open surgery involves a large incision to access a surgical site, whereas minimally invasive surgery involves fewer and smaller incisions, through which a small camera and narrow surgical instruments are inserted to gain access to the surgical site. The first-developed and more traditional minimally invasive surgery, laparoscopy, has an advantage over open surgery in that the smaller incisions can reduce bleeding and lead to faster patient recovery.

The second type of minimally invasive surgery is robotic surgery or robot-assisted surgery, often advertised as one of the most innovative and successful technological advances in recent medical history. As a result of advances in robotics technology, Intuitive Surgical introduced the world's first minimally invasive robotic surgery system, the da Vinci Surgical System, which was approved by the U.S. Food and Drug Administration (FDA) in 2000. In this system, surgical robots provide a magnified high-definition 3D view of the surgical site and enable surgeons to perform surgery by controlling robot arms, which allows for more flexibility and a greater range of motion than is possible with human hands alone. Since its introduction, the da Vinci Surgical System has become the most widely adopted robotic surgical system in the world and has been approved by the FDA for many procedures in urology, thoracic surgery, general surgery, and gynecology, among others. Given its popularity, the system has been installed in more than 1,500 U.S. hospitals, whose estimated annual procedures performed with the da Vinci Surgical System increased from 136,000 in 2008 to 664,000 in 2017 (Childers and Maggard-Gibbons 2018).

Despite its prevalence, the controversy around the value of robotic surgery remains. On the one hand, advocates of robotic surgery argue that superior dexterity and smaller incisions translate to better patient outcomes (Bhama et al. 2016; Lopez et al. 2016; Paek et al. 2016). On the other hand, robotic surgeries have often been criticized for being substantially more costly than laparoscopic surgery without clear evidence of improvement in treatment outcomes, thereby contributing to higher health care costs (Wright et al. 2012). This tension points to the need to determine the true value of robotic surgery for the medical community and policy makers. For health care marketers, a relevant question is whether robotic surgery DTCA is effective in swaying patients to choose robotic surgery over other types of procedure. Understanding the effects of marketing promotion on the basis of consumer responses can provide insights into the debate surrounding the use and promotion of robotic surgery.

In general, robotic surgery has gained popularity as an alternative to the other type of minimally invasive surgery—laparoscopy—because the medical indications for both types of minimally invasive surgery are similar. However, for prostatectomy, robotic surgery has replaced open surgery because the traditional laparoscopy is not an appropriate alternative due to the complexity of the procedure. Thus, in this study, the alternative

surgery type to robotic surgery is open surgery for prostatectomy; for all other organs, laparoscopy is the alternative to robotic surgery for surgical operations.

This study examines the relationship between the level of television DTCA of robotic surgery and patient choice of robotic surgery over alternative types of surgery, an area that has not been studied. In the era of health care consumerism in the United States, numerous studies have found that advertising affects consumer choices in other health care contexts, such as prescription drugs, health insurance policies, and provider choice. However, whether a similar effect prevails regarding the choice of surgical procedure type cannot be answered easily based on this previous work. This is because the choice of surgery type often entails the consideration of more serious health risks compared with the choice of prescription drugs or insurance, suggesting that patients may rely more on physician recommendations and less on advertising when making a choice. Nonetheless, there is anecdotal evidence that some patients specifically request robotic surgery when discussing treatment options (Healthline 2015). This is similar to findings in the DTCA literature that show how patient requests affect physician drug prescriptions (Narayanan and Manchanda 2009; Pareek, Liu, and Ghosh 2019; Stremersch, Landsman, and Venkataraman 2013). Our study takes a first step toward understanding the influence of advertising on the choice of surgery type by estimating the relationship between the choice of robotic surgery and hospital advertising that promotes this procedure.

We gathered quarterly inpatient hospital discharge data between January 2011 and September 2015 (19 quarters) for all hospitals in Florida. These patient-level data include patient demographic information, detailed diagnoses and procedures (robotic surgery or other types of surgery), and zip code of residence. The data also contain detailed information on all hospitals as well as how much the hospital charges payers (e.g., insurance) for each patient visit. In addition, we collected hospital television advertising expenditures and creatives data in Florida during the corresponding period, from which we can identify whether each ad creative promotes robotic surgery.

Some important challenges exist in identifying the causal effect of advertising, as hospitals may endogenously decide on the level of advertising. One approach to identifying the effect would be Shapiro's (2018) border identification strategy, using border counties that take advantage of discontinuity in television advertising across designated market area (DMA) borders. To more accurately measure the causal effects of advertising, we modify this approach to focus on patients who reside within 20 miles of either side of a DMA border in Florida. Specific to our context, supply-side drivers (e.g., the adoption of surgical robots or hospital management directives) can also exist that may be coordinated with robotic surgery advertising, which would then bias our estimate of patient response to advertising. We address this concern using various fixed effects chosen with domain knowledge.

We find a positive and statistically significant impact of robotic surgery advertising on robotic surgeries. In fact, every additional \$1 per 100 capita spent on robotic surgery advertising

during a quarter increases the likelihood of the procedure being chosen by 13.9%, which implies an advertising elasticity of .034. We also conduct supplementary analyses suggesting that robotic surgery may be associated with hospital stays that are approximately five hours shorter.

The results have implications for several stakeholders. First, shorter length of stay (LOS) at the hospital is a clear benefit to patients. As long as the patient's out-of-pocket cost is low (e.g., no or low deductible), robotic surgery ads can improve consumer welfare. We find that Medicaid patients, whose socioeconomic status tends to be low, are more responsive to robotic surgery advertising. Health care providers can better allocate marketing resources and efficiently reach consumers using such customer heterogeneity information. We also find that patients in this market respond only to specific, relevant advertising content. This finding provides an important insight into designing ad content. Our results also show that surgical robot manufacturers can benefit from investing in cooperative advertising with hospitals or engaging in "ingredient branding" (e.g., Intel ads in business-to-business [B2B] marketing) by advertising directly to potential patients.

Our study offers several contributions. First, we provide new insights into the impact of marketing in the health care sector regarding the patient choice of a high-tech surgical procedure for treatment. Although the literature has tended to focus on detailing and DTCA of pharmaceutical products in general, the patient segment requiring surgery is likely substantially different from those who need pharmaceutical drugs. With hospital DTCA having increased more than five-fold in the past two decades, empirically quantifying the effect of hospital marketing promotions on patient choices is important and timely.

Second, we find evidence that consumers in this market respond to the content of the advertisements. Our interviews with hospital marketing departments revealed that they deploy different advertisements promoting various procedures to improve their hospital brand image. These promotions are based on requests from different specialties, despite the lack of clear evidence of a return on investment for such advertisements. We find that patients respond to robotic surgery ad creatives when choosing the type of surgery but do not respond to ad creatives without robotic surgery content. In addition, we find customer heterogeneity in response to advertising in terms of insurance status. Given these results, marketers should consider specific advertisement content carefully when they develop marketing plans for the health care industry.

Third, using fixed effects and matching methods, we find evidence that robotic surgery is associated with both shorter LOS and higher costs. Combining this with our finding that advertising sways patient choice of surgery type, our study suggests that ads can hurt insurance providers that offer plans with low out-of-pocket costs, especially when the benefit of shorter LOS is deemed minimal. In addition, regulators might consider requiring more transparency in high-tech procedure ads so that consumers are better informed about the pros and cons of the procedure as well as associated costs.

The rest of the article is organized as follows. We first provide an overview of the relevant literature, followed by a discussion of the industry background on robotic surgery. We then describe the data and the econometric models we estimate and provide the results on the effect of advertising on the choice of robotic surgery. We then explore advertising content effects and heterogeneous advertising effects. To gain further insights, we also report the results from a supplementary study on the effect of robotic surgery on patient health outcomes and costs. We conclude with a general discussion and implications for public policy and management.

Relevant Literature

Given the large size of the U.S. health care market, numerous studies, as we have previously discussed, have examined marketing practices directed at consumers and patients. The most closely related literature to the focal study is on advertising: Aizawa and Kim (2018) and Shapiro (2020) examine health insurance advertising for Medicare Advantage plans, while Karaca-Mandic et al. (2017) and Aizawa and Kim (2020) investigate the insurance industry and government advertising promoting the Affordable Care Act rollout. However, advertising focused on hospital care has been understudied with the exception of Kim and KC (2020b), who investigate the effect of general hospital advertising on attracting patients in Massachusetts. Our study develops this line of literature by showing that high-tech surgical procedure advertising can affect the choice of treatment options in hospitals that engage in this promotion. In addition, our results show that advertising content focused on a certain medical procedure plays a role.

High-tech procedures and high-tech medical services have been the center of the “medical arms race” argument, which suggests that hospital adoptions of high-tech medical services have led to increased costs and wasteful hospital competition (Cutler and McClellan 1996; Newhouse 1992). However, empirical studies find contrary results showing that high-tech procedures and equipment are mainly limited to highly concentrated markets (Dranove, Shanley, and Simon 1992; Luft et al. 1986). Testing the “medical arms race” argument is outside the scope of our study. Instead, we unpack the demand for high-tech medical procedures by investigating the role of advertising in this market. Studies have also looked at other technology adoption in hospitals (e.g., magnetic resonance imaging adoption in Baker [2001], electronic medical records in Miller and Tucker [2009, 2011]). Our data sample starts more than a decade after the FDA approval of the surgical robot; therefore, the technology was already available in the hospitals in our data. Thus, rather than studying hospital adoptions, we focus on the effort to increase the utilization of the technology through advertising.

Our study is also related to the literature examining the value of robotic surgeries. Results from medical studies of robotic surgery are mixed, with some documenting its short-term benefits, such as shorter LOS (Bhama et al. 2016; Lopez et al. 2016) and less bleeding (Paek et al. 2016). Other studies (Iavazzo,

Mamais, and Gkegkes 2016; Jayne et al. 2017; Jeong et al. 2017), however, provide no strong evidence that robotic surgery is superior to conventional laparoscopic surgery. Each of these medical studies, whether favorable or unfavorable toward robotic surgery, typically looks at one specific robotic surgery procedure (e.g., hysterectomy, colectomy, nephrectomy) or robotic surgeries in one medical department (e.g., colorectal surgery, gynecology, urology). Our study differs in that we examine the medical value of robotic surgery for various types of robotic surgery procedures across five medical departments and thus report a more extensive comparison of robotic surgery and conventional laparoscopic surgery based on a large patient population. In addition, unlike research on DTCA of prescription drugs, our context and data allow us to measure short- and long-term health outcomes, such as bleeding, LOS, hospital readmission, in-hospital mortality, and two-year mortality. These rich outcome measures, combined with hospital charge data, provide further insights into the implications of robotic surgery advertising and raise the issue of consumer protection in medicine. In the medical marketing literature on consumer protection, Yoon (2020) documents the detrimental impact of publishing cardiac surgeon report cards on urgent patient death rates after heart surgery. Our study adds to the literature by showing that advertising high-tech medicine may lead patients to choose a high-tech procedure whose surgical and economic advantages compared with conventional alternatives are still being debated.

Industry Background

Patient Surgery Decisions

In this section, we outline the patient surgery decision process and define the appropriate scope of research in studying the effectiveness of robotic surgery advertising.

Surgical versus medicine treatment. Many patient ailments or diseases can be treated with either medicine or surgery. For many diseases, noninvasive treatment (i.e., medicine) is naturally the first line of care before considering riskier surgical treatment, which may result in severe complications. Surgery is an option only for patients who do not respond to medicine well or who have conditions that can only be treated surgically. For instance, upon diagnosis of certain types of disease, such as benign tumors or early-stage malignant tumors, surgical treatment is the first treatment of choice to remove localized tumor cells or diseased organs. This is particularly true for patients whose surgery options include robotic surgery. In fact, more than 90% of robotic surgery procedures in our data are for resection of the prostate, uterus, cervix, ovary, lung, kidney, stomach, bowel, gall bladder, or hernia. For example, for those who need a hysterectomy (i.e., removal of the uterus), medicine treatment options are not relevant at the point of surgery decision. Therefore, the relevant patient population in relation to robotic surgery advertising would be those who need surgery because medicine treatment is ruled out or has failed. Thus,

we do not consider the choice between surgical and medicine treatments.

Surgery type choice versus surgery incidence. An important question regarding advertising for robotic surgery is whether it has led to more surgeries similar to the market expansion effects of pharmaceutical DTCA. The literature on DTCA has shown that it is effective at expanding the market through different mechanisms, such as prompting more patient visits (Iizuka and Jin 2005) and spilling across brands in a category (Liu, Steenburgh, and Gupta 2015). However, we do not expect similar effects of DTCA for surgical care and abstract away from modeling the incidence of surgery for several reasons. First, robotic surgery advertising content is generally different from pharmaceutical DTCA. Pharmaceutical DTCA often describes possible symptoms and shows people suffering from certain conditions, which can increase awareness of those health concerns. In addition, these advertisements often encourage the audience to ask doctors about the featured condition, which can account for increases in patient visits and lead to market expansion for pharmaceutical drugs. In contrast, robotic surgery advertising does not mention specific symptoms, but prominently displays the robotic surgery system. There is also little possibility that spilling across brands in the surgery category would occur, as there is no other prominent brand in robotic surgery other than da Vinci Surgical Systems. Taken together, it is unlikely that robotic surgery advertising leads to market expansion of surgery.

Second, to empirically test this effect, one would need survey data with enough observations of patients who decline necessary surgery. However, the hospital discharge data in our study do not include patients who decline surgery because they are not admitted to hospitals.

Third, studies examining patient choice regarding hospital-based invasive surgery assume that patients seek surgical care only when necessary (for further discussions, see Gaynor, Propper, and Seiler [2016] and Kim and KC [2020b]). In other words, a patient's decision is limited to the choice of surgical care, and we abstract away from the incidence of surgery. Therefore, we examine the patient's choice of surgery type instead of surgery incidence.

Robotic surgery and its alternatives. As discussed previously, three types of surgical procedures are performed: open surgery and two types of minimally invasive surgery, namely, laparoscopic surgery and robotic surgery. Because both robotic surgery and laparoscopy are minimally invasive surgeries, the patients who undergo such procedures are similar, as are their medical conditions. Interviews with physicians and surgeons revealed that laparoscopy is a substitute for robotic surgery, but open surgery is not. The only exception here is prostatectomy, where open surgery, not laparoscopy, is the alternative to robotic surgery (Web Appendix A for details).

Hospital choice versus surgery type choice. Advertising may affect a consumer's choice of a hospital as well as the type of surgery.

We argue that it is reasonable to assume a sequential decision of choosing the hospital before the type of surgery, given the specifics of the typical patient journey in the U.S. health care system. In addition, we do not find empirical evidence that robotic surgery advertising affects market shares. Therefore, we abstract away from the hospital choice and focus on the choice of surgery type. In other words, the proper interpretation of the advertising effect in our model is the choice of robotic surgery over the alternative surgery at a chosen hospital (see Web Appendix B for further discussions).

Television Advertising Market

In the United States, television advertisers purchase local advertising spots at the DMA level. AC Nielsen delineated DMAs in 1955; each DMA comprises a group of counties and is constructed so that cable or satellite television consumers within the DMA receive the same television content and advertising (for more details on DMAs and localized over-the-air signals, see Shapiro [2018]). In addition to local advertising spots, advertisers can purchase advertising at the national level, covering all U.S. television markets; our data allow us to count national advertisement spending separately from local advertisement spot spending. Unlike consumer packaged goods or prescription drugs, hospital demand is locally concentrated, which means that hospitals typically focus on local television advertising (i.e., they advertise in the DMA where they are located). Hospitals in our data sample engage in local advertising only (no national advertising). Florida contains ten DMAs: Mobile-Pensacola, Panama City, Tallahassee-Thomasville, Jacksonville, Gainesville, Tampa-St. Petersburg, Orlando-Daytona Beach-Melbourne, Ft. Myers-Naples, West Palm Beach-Ft. Pierce, and Miami-Ft. Lauderdale. We exploit advertising variation across the DMAs and time to study the effect of robotic surgery advertising on the type of surgery chosen.

Data

In this section, we describe our data sets and provide summary statistics.

Inpatient Data

We gather quarterly individual-level inpatient discharge records from hospitals in Florida between January 2011 and September 2015 (19 quarters). Note that our hospital discharge data show only patients who visited the hospital for admissions, and we typically observe one hospital admission per patient for surgery in our data, as patients requiring multiple surgeries during our five-year data sample are rare. Each visit record shows the hospital from which a patient is discharged in a given quarter, along with detailed patient demographic factors (e.g., gender, age) and insurance information (e.g., Medicare, Medicaid, private insurance). These data also include detailed diagnosis and procedure codes based on the International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9), a standard that

includes over 14,000 diagnosis codes and 4,900 procedure codes. We identify the type of surgery (robotic, laparoscopic, or open surgery) using ICD-9 codes (see Web Appendix C.1 for detailed decoding schemes). We parsimoniously capture patient health conditions by identifying their level of comorbidities. Comorbidity refers to the presence of two or more health conditions in a patient; a higher comorbidity index is associated with more complex clinical management and worse patient outcomes. We extract 31 comorbidity measures (e.g., cardiac arrhythmia, hypertension, diabetes, metastatic cancer, obesity) from ICD-9 diagnosis codes using the Elixhauser Comorbidity Index (Elixhauser et al. 1998) and count the number of comorbidities present in a given patient visit.

The inpatient data also contain the five-digit zip code of each patient's primary residence. We match 2010 median household income data from the U.S. Census based on the zip code as a proxy for each patient's income. The zip codes are also matched to the DMAs of local television advertising, which reveals the level of advertising for each patient.

Because we are interested in the marginal effect of robotic surgery advertising, we identify disease categories for which robotic surgery and its alternative (i.e., open surgery for prostatectomy and laparoscopy for all others) are substitutable as surgical options. Our disease categories are defined by using the single-level Clinical Classifications Software developed as part of the Healthcare Cost and Utilization Project. We choose disease categories in which both the number of robotic surgeries and the number of alternative surgeries equal at least 50 across all hospitals in Florida in our data period (19 quarters). In other words, our sample excludes disease categories where there is no substitution for a surgery type because one type of surgery is the usual standard of care and another type of surgery is rarely performed. In the final sample, we identify five medical departments: gastrointestinal surgery, hepatopancreatobiliary surgery, thoracic surgery, obstetrics and gynecology (ObGyn), and urology. This data cleaning process leaves us with 139,885 patient visit records in 56 disease categories across five departments from 121 hospitals in Florida. Note that the 56 disease categories account for 97.5% of all robotic surgery procedures in Florida, and all but four hospitals conduct robotic surgery in the sample.

By tracking patient admissions and decoding the ICD-9 codes, we also extract various patient outcome measures, such as in-hospital mortality, two-year mortality, 30-day hospital readmission, LOS, and bleeding. In addition, we collect charge data for each visit. These data record the charges that hospitals and providers bill to insurance companies (or to patients with no health insurance). The charge amounts in the data may be different from the actual payments that providers receive. Even so, because actual revenues are more likely to be larger when billed amounts are larger, the charge data are useful for understanding which type of surgery costs more. Summary statistics of health outcomes and charges are included in Web Appendix C.2.

We detail descriptive statistics of our inpatient data sample in Panel A of Table 1. Patients are, on average, 56.85 years old, and 60% are female. The average median household income is

approximately \$51,350. With regard to race and ethnicity, 66% are white, 12% are Black, 18% are Hispanic, and the remaining 4% are of other ethnicities. The two largest government-run insurances cover nearly half of the patients (41% Medicare and 7% Medicaid), followed by 45% covered by private insurance and 8% self-paying or uninsured. The average number of comorbidities, a proxy for health conditions, is 1.75. Panel A also reports the summary statistics for the same patient characteristic variables according to surgery type. The proportion of female patients undergoing robotic surgery (51%) is lower than that of those having the alternative (65%) because robotic surgery is the most popular in the urology department, whose patients are primarily male. The Comorbidity Index suggests that robotic surgery, rather than nonrobotic surgery, is typically performed for relatively healthier patients.

Table 1, Panel B, shows the number of surgical cases by surgery type and by medical department. Robotic surgery in the urology department is dominant because more than 70% of urologic minimally invasive surgeries are for prostate cancer, and robotic prostatectomy is the standard surgical treatment. Robotic surgery is also a major minimally invasive procedure in the ObGyn department, where this surgery is used for most uterine tumors. In other departments, robotic surgery is increasingly used for colorectal cancers, gallbladder stones, and lung cancers.

Hospital Advertising Data

We obtained advertising data from Kantar Media to track the monthly television advertising expenditures of each hospital in a DMA in Florida. To determine the amount of hospital advertising, we scale the advertising level to dollars per 100 capita using yearly DMA population data from the U.S. Census Bureau. Since the relative intensity of advertising spending in a market depends on its population size, this scaling effectively normalizes ad spending across DMAs with different population sizes, similar to Shapiro (2018) and Kim and KC (2020a, b). We then aggregate advertising expenditures to the quarterly level to match them with our quarterly inpatient data. As in Sinkinson and Starc (2019), our measure of advertising is the average of the advertising levels in the discharge quarter and the previous quarter. Although previous research has shown a lagged advertising effect (e.g., Dubé, Hitsch, and Manchanda 2005), we parsimoniously capture it using the average for two reasons. First, Kim and KC (2020b) show that the effect of general hospital advertising in the current month depreciates to below 10% of its magnitude after four months, suggesting that lagged advertising beyond the previous quarter may have few residual effects. In Web Appendix E.1, we show that the advertising carryover effect does not seem to prevail in our quarterly data. Second, we need to account for the delay between advertisement impression and hospital admission. The average 3.4-day (SD = 4.41 days) hospital stay until discharge implies that our quarterly data based on patient discharge dates may not be precise enough to estimate advertising persistency even with functional form assumptions.

Table 1. Inpatient Data Summary Statistics.**A: Patient Characteristics**

	All					Robotic Surgery		Nonrobotic Surgery	
	N = 139,885					N = 51,522		N = 88,363	
	Mean	SD	p10	Median	p90	Mean	SD	Mean	SD
Demographics									
Age (years)	56.85	15.85	36	58	76	58.72	13.92	55.77	16.77
Female	.60	.49	0	1	1	.51	.50	.65	.48
Income (\$000)	51.35	15.70	34.52	48.63	71.39	51.85	15.97	51.05	15.53
Race and Ethnicity									
White	.66	.47	0	1	1	.64	.48	.67	.47
Black	.12	.33	0	0	1	.12	.33	.12	.33
Hispanic	.18	.38	0	0	1	.19	.40	.17	.37
Other	.04	.20	0	0	0	.05	.21	.04	.19
Insurance									
Medicare	.41	.49	0	0	1	.41	.49	.40	.49
Medicaid	.07	.25	0	0	0	.05	.22	.08	.26
Private	.45	.50	0	0	1	.48	.50	.44	.50
Other	.08	.26	0	0	0	.06	.23	.09	.28
Medical Condition									
Comorbidity index	1.75	1.65	0	1	4	1.53	1.53	1.87	1.70

B: Number of Surgical Cases by Department

Department	Robotic Surgery		Nonrobotic Surgery		Total
Gastrointestinal Surgery	10,039	15.3%	55,700	84.7%	65,739
Hepatopancreatobiliary Surgery	885	7.8%	10,402	92.2%	11,287
Thoracic Surgery	5,070	40.7%	7,391	59.3%	12,461
Obstetrics and Gynecology	14,419	59.2%	9,943	40.8%	24,362
Urology	21,109	81.1%	4,927	18.9%	26,036
Total	51,522	36.8%	88,363	63.2%	139,885

Notes: Nonrobotic surgery includes open surgery for partial or complete removal of the prostate (open prostatectomy) and laparoscopy for other procedures. The comorbidity index accounts for the number of Elixhauser Comorbidity conditions based on the diagnosis codes. p10 and p90 denote the 10th and 90th percentiles, respectively.

Therefore, we take the average of the advertising levels in the discharge quarter and the previous quarter for our advertising measure.

Interviews with hospital marketing departments revealed that they deploy different advertisements promoting various procedures (e.g., one advertisement promoting its cancer center and another one promoting robotic surgery), often based on requests from different medical specialties. A vice president of marketing at a large health care system in Florida told us that, in addition to the marketing department's own advertising agenda and schedule, doctors often request a meeting with the marketing department to run promotions of their own specialty and deliver a specific message (e.g., the availability of robotic surgery). If their advertising schedule allows, the hospital marketing team contacts an advertising agency with the specified messages and budget. This latter decision is ad hoc and, not surprisingly, depends on the status of the requesting doctor. For instance, if a department in a given hospital has a world-renowned surgeon and the

surgeon requests a meeting to discuss advertising, it is difficult to turn this request down. This practice suggests that hospital advertising may not always be targeted and planned, implying that endogeneity issues may not be severe in our data.

We also collected television advertising video content from Kantar Media to determine which ad creatives promote robotic surgery. Hospital advertising typically promotes general hospital care, emergency care, cardiac care, cancer patient care, women's care, and specific surgical care, among others. We distinguish robotic surgery ad creatives from others by reviewing the ad creative titles and transcriptions of the video content and confirm them by watching the individual videos. This process resulted in 25 ad creatives from 15 hospitals (see Web Appendix C.3.1 for further details on the data collection process).

Note that due to limited marketing budgets and the locality of the patient pool, hospitals in our sample engage only in local spot television advertising (i.e., only in the DMA where they are located). Across these 15 hospitals, spending on television

advertising of robotic surgery in the market (i.e., in their own DMAs) is \$271,505 during the study period, which accounts for approximately 6.4% of their total spending (\$4,238,846) on television hospital advertising. We observe substantial temporal variation in advertising across hospitals and time (Figure 1 and Figure W1 in Web Appendix C.3.2). As shown in Table W5 in Web Appendix C.3.3, the average spending of the 15 hospitals on robotic surgery advertisement is \$.049 per 100 capita per quarter (SD = \$.252 per 100 capita per quarter), while their spending on all types of hospital advertising is \$.790 per 100 capita per quarter (SD = \$1.731 per 100 capita per quarter). The average ad spending of these hospitals is higher than that of all hospitals' spending (\$.532 per 100 capita per quarter) in Florida.

We also find that the proportion of robotic surgery in these 15 hospitals (38.9%) is higher than that in the hospitals that do not advertise robotic surgery (36.7%). We further show descriptive evidence on the effect of robotic surgery advertising on the choice of surgery mode in Web Appendix F.

Econometric Models

This section outlines econometric models used to examine the effects of advertising on the choice of robotic surgery. We also specify the model to examine the choice of robotic surgery on health outcomes and costs.

Choice of Robotic Surgery and Advertising

We first explain our identification strategy using the DMA borders to address the potential bias of advertising effects. We then present our main model, which uses border samples.

Border identification. We modify the DMA border identification strategy developed in Shapiro (2018) to more accurately measure the causal effects of advertising. Since the level of hospital television advertising varies across DMAs, the traditional setup is to use adjacent counties around DMA borders, considering one side of the border the control group and the other side the treatment group, as in difference-in-differences estimation. This identification is based on the assumption that advertisers typically focus on areas with a large customer population, which are often located at the center rather than the outskirts, border areas of a DMA. A direct application of Shapiro's DMA-county-border strategy may not be appropriate in Florida, because counties adjacent to DMA borders in Florida include large population centers due to the unique geographic shape of the state (see Web Appendix D.1 for detailed discussion). To overcome this issue, we construct our own DMA border areas that are more granular than counties by focusing on patients residing within 20 miles of a DMA border (20 miles on one side of a border and 20 miles on the other side). Our data sample then consists of patients whose residences are in the DMA border regions shown in Figure 2, where circles and their size represent zip-code-level patient locations and the corresponding patient volume.

Note that a tighter distance bandwidth around the DMA border increases the internal validity of the process of identifying

the effect of advertising, but it raises two important issues. First, given that we focus on DMA border regions in Florida, our results may not be generalizable beyond our sample. To address this concern, we conduct a separate analysis using the entire population of Florida with various fixed effects and find robust results (see Table 2). Second, the use of a very narrow bandwidth may make the advertising effect estimate less precise because the border sample for the estimation may have too few patients for statistical inference. To address this issue, similar to the use of mean squared errors to choose an optimal bandwidth in regression discontinuity methods (Imbens and Lemieux 2008; Hartmann, Nair, and Narayanan 2011), we test different bandwidths (from 5 miles to 40 miles by 5-mile increments) and find that using the 20-mile bandwidth leads to the smallest mean squared error. This gives us confidence that the 20-mile bandwidth in our main model is appropriate.

There are other nuanced differences between our identification strategy and the conventional border identification strategy. In our case, a hospital runs TV advertising in its own DMA. Therefore, for the hospital, we observe local (distant) patients in the same (adjacent) DMA who are (are not) exposed to robot surgery advertising. We assume that if there is no robotic surgery advertising, the market share of robotic surgery among local patients would be predicted by the market share of robotic surgery among distant patients after controlling for various factors. Even though the assumption is not testable, we believe that it is a reasonable assumption for a narrow band of areas (20 miles) around DMA borders. We provide robustness checks, along with detailed comparisons of the identification strategy across several dimensions (data structure, geographic coverage of advertising, and product availability), in Web Appendix D.3.

We consider other potential threats to identification, such as measurement errors related to patients crossing DMA borders to visit hospitals, unobserved patient characteristics related to distances to the hospital, and unobserved preferences of patients to seek emergency care. The measurement error is minimal given that we observe each patient's zip code of residence. We also control for the patient's distance from the hospital and find robust results. For emergency patients, we conclude that their preferences for closer hospitals would underestimate our ad effects (i.e., a conservative measure). Further details can be found in Web Appendix D.4.

Main model. We use Equation 1 to measure the causal effects of robotic surgery advertising on the choice of robotic surgery:

$$\text{Robot}_p = \beta_0 + \beta_1 \text{RobotAd}_{c_p h_p t_p} + X_p \gamma + \delta_{h_p d_p b_p c_p} + \lambda_{h_p d_p b_p t_p} + \varepsilon_p. \quad (1)$$

Our dependent variable, Robot_p , equals one, for patient p , if robotic surgery is chosen, and zero if laparoscopic surgery (or open surgery for prostatectomy) is chosen. $\text{RobotAd}_{c_p h_p t_p}$ is the amount of advertising spending per capita of hospital h on robotic surgery in the patient's county (c_p), determined by patient p 's zip code of residence. Note that h_p is the hospital where the patient p is discharged at time (year-quarter) t_p . The parameter of interest is β_1 . A positive and statistically significant

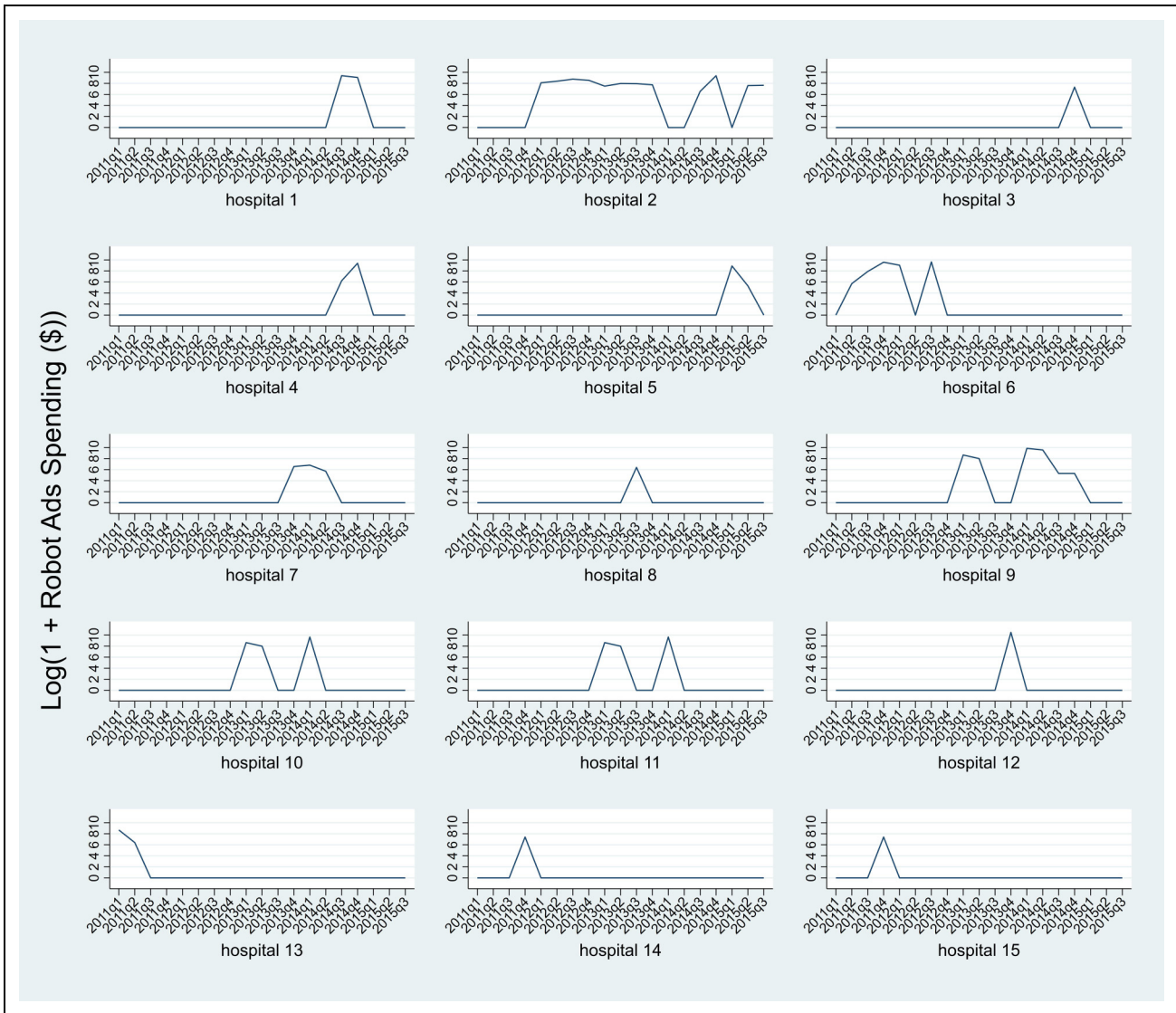


Figure 1. Hospital Advertising on Robotic Surgery over Time.

β_1 would suggest that hospital robot advertising increases the likelihood of choosing robotic surgery over the alternative surgery type. X_p is a set of patient risk factors and characteristics such as demographics, ethnicity, health insurance, and a proxy for health conditions.

Border identification takes advantage of a quasiexperiment for each border and in each quarter and requires two sets of fixed effects: (1) border-county fixed effects to control for spatial heterogeneity in choosing robotic surgery and (2) border-time fixed effects to control for the common time trend across the border. To account for several other factors, we need more granular fixed effects in our model. First, each of these fixed effects needs to vary by the hospital because we measure the choice of surgery type within a hospital, as discussed in the “Industry Background” section.

Second, fixed effects need to vary by medical department. Even in the same hospital, patient preferences for robotic

surgery can differ among medical departments because each department has patients with different medical conditions and surgeons with varying levels of familiarity and skill in robotic surgery.

Third, hospitals may decide to increase robotic surgery utilization by purchasing new robotic surgery systems or offering hospital admitting and surgical privileges to surgeons who are more familiar with robotic surgery. If this decision is accompanied by a marketing promotion, the advertising effect may be overestimated. More importantly, the increase in robotic surgery utilization due to personnel changes varies by the medical department. For example, a urology department giving hospital admitting privileges to a surgeon who is familiar with robotic surgery would not increase robotic surgery utilization in the obstetrics and gynecology department.

Consequently, we include two sets of fixed effects in Equation 1: hospital-department-border-county fixed effect

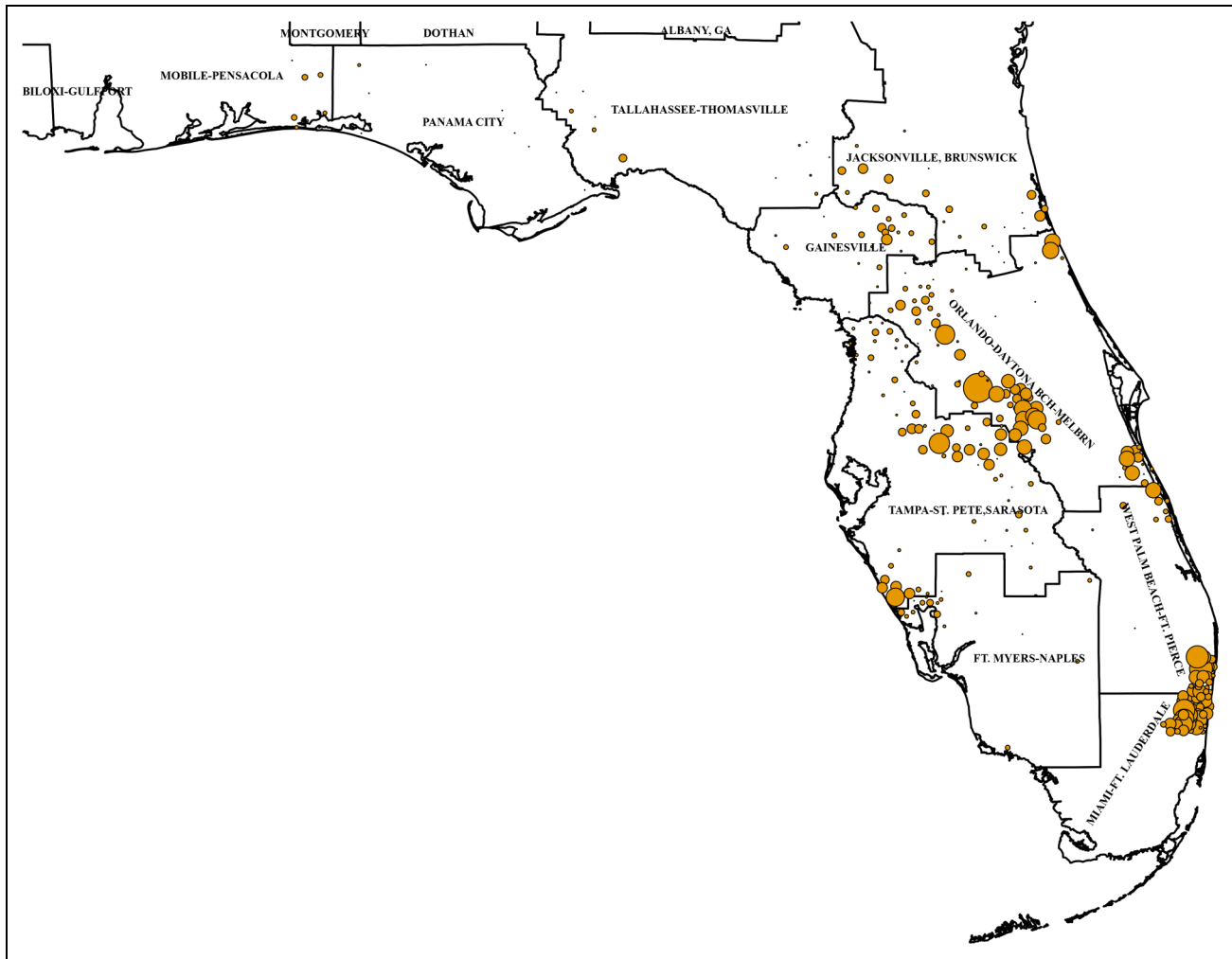


Figure 2. DMA Border Sample.

Notes: The base layer of Figure 2 shows DMAs in Florida. Circles represent patient locations within the 20-mile bandwidths from DMA borders, with their size representing patient volume.

$\delta_{h_p d_p b_p c_p}$ and hospital-department-border-time fixed effect $\lambda_{h_p d_p b_p t_p}$. Here, d_p indicates the medical department where patient p has surgery, and b_p denotes the border determined by patient p 's zip code of residence. Including hospital and department fixed effects, we measure the advertising effect in the residuals after hospital and department choices. Therefore, in our model, the proper interpretation of the advertising effect is the choice of robotic surgery over the alternative surgery type for patient p 's visit at department d of the chosen hospital h . A potential concern is that these granular fixed effects may leave little data variation in $\text{RobotAd}_{c_p h_p t_p}$. We address the concern by further discussing the remaining data variation after the inclusion of fixed effects and showing residual variation in Web Appendix E.4.

As described previously, our main model applies a border identification strategy using the sample of patients residing within 20 miles of a DMA border. As a comparison, we also estimate parameters using the full sample model in which we

use hospital-department-county fixed effects ($\delta_{h_p d_p c_p}$) and hospital-department-time fixed effects ($\lambda_{h_p d_p t_p}$) instead of border-level fixed effects. We estimate parameters using a linear probability model and check the robustness of the nonlinear models in Web Appendix E.2.

Patient Outcomes and Costs

If robotic surgery advertising affects patient choice of robotic surgery, this raises the question as to whether a change in choice has implications for patient outcomes and costs. Ideally, a randomized control trial (RCT) would need to be conducted in which patients are randomly assigned to either robotic surgery or the alternative surgery type, and then any differences in the outcomes between the two groups are measured. However, conducting an RCT with a large number of patients across various departments is not feasible due to ethical and public health concerns. Instead, we conduct supplementary analyses to

Table 2. Parameter Estimates of Main Model And Border Identification.

Model Sample	(1) Full	(2) Full	(3) Full	(4) 20-Mile Border
Robot ads (\$/100 capita)	.037* (.019)	.048** (.019)	.121*** (.008)	.139*** (.026)
Demographics				
Age		.004*** (4.68×10^{-4})	.001*** (2.96×10^{-4})	.001*** (2.99×10^{-4})
Female		-.095*** (.022)	-.013*** (.004)	-.008 (.005)
Income (\$000)		-1.55×10^{-4} (1.70×10^{-4})	-4.93×10^{-5} (9.64×10^{-5})	7.62×10^{-5} (1.17×10^{-4})
Black		.055*** (.010)	-.004 (.005)	.004 (.006)
Hispanic		.017 (.011)	.008 (.005)	.004 (.008)
Other ethnicity		.057*** (.011)	.008 (.006)	.007 (.011)
Insurance				
Medicaid		.007 (.023)	-.018** (.009)	-.017 (.012)
Private insurance		.058*** (.008)	.011* (.006)	.005 (.010)
Other insurance		.002 (.013)	.001 (.008)	-.024** (.012)
Medical Condition				
Comorbidity Index		-.032*** (.002)	-.005*** (.001)	-.005** (.002)
Hospital FEs	Y	Y		
Hospital-department-county FEs			Y	
Hospital-department-border-county FEs				Y
Quarterly FEs	Y	Y		
Hospital-department-quarterly FEs			Y	
Hospital-department-border-quarterly FEs				Y
Number of observations	139,885	139,885	139,885	43,700
R ²	.135	.164	.542	.596

* $p < .1$, ** $p < .05$, *** $p < .01$.

Notes: FEs = fixed effects; Y = yes. The dependent variable represents a dummy variable that equals one if robotic surgery is chosen, and zero if laparoscopic surgery (or open surgery for prostate) is chosen. The reference ethnicity is white, and the reference insurance is Medicare. Robust standard errors clustered at the hospital level are reported in parentheses. In Column 4, the number of hospital-department-border-county fixed effects is 1,510, and that of hospital-department-border-quarterly fixed effects is 5,495. In this sample, there are 119 hospitals, 5 medical departments, 23 DMA borders, and 86 patient residence counties during 19 quarters. Web Appendix E.5 shows that model specifications with less granular fixed effects generate consistent results.

identify any evidence of different outcomes and costs for the types of surgery in our data. Our data allow us to compare a large number of patient outcomes (LOS, in-hospital mortality, two-year mortality, 30-day hospital readmission, and bleeding) and costs of two surgery types, robotic surgery and the alternative (open surgery for prostatectomy or laparoscopy for the other procedures). We use Equation 2 to examine the relationship between robotic surgery and its patient outcomes and costs.

$$\text{Outcome}_p = \alpha_0 + \alpha_1 \text{Robot}_p + W_p \theta + \xi_{hp} + \tau_p + \omega_p. \quad (2)$$

Outcome_p captures health outcomes or costs for a patient p. α_1 captures the effect of robotic surgery on the outcomes

for all hospitals. If the outcome of interest is LOS and α_1 is negative (positive), then robotic surgery is associated with a shorter (longer) LOS. W_p denotes a set of patient characteristics and risk factors such as age, gender, income, ethnicity, insurance, the Elixhauser Comorbidity Index, and dummy variables for five departments, which allows us to capture risk factors affecting outcomes that are inherent to the type of patients and the group of diseases varying by the medical department. We also include hospital fixed effects (ξ_{hp}) to account for hospital-specific factors that may affect outcomes. Lastly, because outcomes related to robotic surgery can improve nonlinearly during our data sample as the technology in the field matures, we include quarterly

fixed effects τ_p to account for the nonlinear progression of the technology in each time period.

Despite the patient control variables and fixed effects included in our model, α_1 may be biased because of other factors that can affect both the outcome and the decision to choose robotic surgery.² To account for this, we check the robustness of our results by addressing selection on observables using two matching methods: propensity score matching (PSM) and coarsened exact matching (CEM) methods (see Web Appendix I.1.1 for details). These matching methods can address selection bias related to observed patient characteristics and surgery characteristics. A finding of consistent results similar to the ordinary least squares (OLS) linear regression model with fixed effects would suggest that our rich covariates may be sufficient to capture the selection of robotic surgery; however, if unobserved patient health conditions affect both the decision for robotic surgery and outcomes even after matching, our estimate may still be subject to bias.

Results

This section presents the main results of the study. We examine the effect of robotic surgery advertising using the full sample with fixed effects and then focus on the 20-mile DMA border sample to address endogeneity concerns that may bias parameter estimates. We then discuss the effects of advertising content, heterogeneous advertising effects, and patient outcomes and costs, as well as various tests for robustness.

Main Advertising Effects on the Choice of Procedure

Main model results. Before using the 20-mile DMA border sample, we first estimate OLS regressions for the full sample with various fixed effects. Columns 1–3 of Table 2 present the results obtained by estimating Equation 1 using the entire patient sample in Florida. Column 1 shows that, after controlling for hospital- and time-specific factors using fixed effects, a greater amount of robotic surgery advertisements seems to marginally increase the likelihood of robotic surgery being chosen (.037, $p < .1$). In Column 2, we control for various patient risk factors such as demographics, health insurance status, and a proxy for health conditions. We find a statistically significant effect of advertising (.048, $p < .05$; 95% CI: [.011, .086]). We include more granular fixed effects in Column 3: hospital-department-county-level fixed effects ($\delta_{h_p d_p c_p}$) and hospital-department-time-level fixed effects ($\lambda_{h_p d_p t_p}$). The advertising effect estimate becomes more positive and more statistically significant (.121, $p < .01$). This suggests that even after considering potential county-level geographic advertisement targeting (subsumed in $\delta_{h_p d_p c_p}$) and supply-side drivers of

hospitals (subsumed in $\lambda_{h_p d_p t_p}$), the advertising effect on the choice of robotic surgery becomes greater (95% CI: [.104, .137]).

To further address any remaining endogeneity concerns regarding advertising that may bias the estimates, we restrict our sample to within 20 miles of a DMA border (i.e., patients who reside within 20 miles on either side of a DMA border, as shown in Figure 2) and use the hospital-department-border-county fixed effect and the hospital-department-border-time fixed effect, as previously discussed. The point estimate of the advertising effect in Table 2, Column 4, becomes .139, statistically different from zero at the 1% significance level (95% CI: [.088, .189]). The point estimate is greater than that of the full sample in Column 3, but these are not statistically different from each other at the 5% significance level. These results imply a causal advertising effect on a patient's choice of robotic surgery over the alternative surgery type in Florida. The parameter estimate suggests that spending an additional \$1 per 100 capita on robotic surgery advertisements increases the likelihood of this procedure being chosen by 13.9% (95% CI: [8.8%, 18.9%]). This estimate implies that a 1% increase in the advertising level increases the probability of choosing robotic surgery by .034%. Note that robotic surgery advertising does not affect a hospital's market share or substitutions between minimally invasive surgery and open surgery, as documented in Web Appendices A and B. The effect is, therefore, likely driven by the substitution between robotic surgery and the alternative surgery type within a hospital.

For parameter estimates of patient characteristics, we focus on the estimates in Table 2, Column 4, from our border identification. The results show that older patients or patients with a small number of comorbidities are more likely to choose robotic surgery over the alternative surgery type. Parameter estimates of gender, income, and ethnicity are not statistically different from zero. Patients with other types of insurance (47% of these patients do not have insurance) are significantly less likely than Medicare patients to choose robotic surgery.

For the remainder of the article, our preferred model is Table 2, Column 4, because our border identification provides more credible estimates. Thus, the remaining results that we present are based on Column 4's specification.

Robustness checks. We report our main findings using a linear probability model for the following reasons. First, as Angrist and Pischke (2009) point out, linear probability models are easy to interpret and produce results similar to those obtained using nonlinear models such as logit and probit. Second, Goldfarb and Tucker (2011) note that estimating a nonlinear model with a large set of individual fixed effects is computationally burdensome. In our setting, a linear probability model allows us to easily incorporate the high-dimensional dummy variables such as hospital-department-border-county fixed effect $\delta_{h_p d_p b_p c_p}$ and hospital-department-border-time fixed effect $\lambda_{h_p d_p b_p t_p}$. One issue with using linear probability models is the out-of-range prediction problem. However, the linear probability model estimation with our sample produces no out-of-range predictions, the range being between .271 and .637. As a robustness check, we show

² Our border identification strategy using the discontinuity of advertising across DMAs is useful to identify the advertising effect; however, it does not overcome selection bias related to patient outcomes.

in Web Appendix E.2 that similar results are obtained with logit or probit models when estimating nonlinear models with fewer fixed effects (e.g., omitting the department-level fixed effects due to the computational burden). As for functional form assumptions of the advertising variable in our model, we find that a logarithmic function of advertising produces qualitatively similar results (see Web Appendix E.3). We also test the carryover effects of advertising and find that the carryover effect of advertising from the previous quarters is small and statistically not different from zero (Web Appendix E.1). Finally, for our border identification strategy using patients within 20 miles of a DMA border, we conduct a series of balance tests and find no discontinuity of patient characteristics across borders that is predicted by the advertising of robotic surgery (Web Appendix D.2). We also vary the bandwidth distance from the DMA borders from 20 miles to 30 and 40 miles and find robust results (Web Appendix E.6).

Advertising Content Effects and Heterogeneous Effects

In this section, we present advertising content effects and heterogeneous effects to broaden our understanding of the effects of advertising in this market.

Response to advertising content. We explore responses to advertisement content. We show that more robotic surgery advertising increases the likelihood of robotic surgery (Table 3). To check that this relationship is driven by robotic surgery advertising content, we conduct a placebo test by replacing robotic surgery advertising expenditures with hospital advertising expenditures without robotic surgery ad content. Column 2 of Table 3 shows a nonsignificant effect, suggesting that advertising content matters in this market. We also check a possible spillover effect of robotic surgery advertising. If other hospitals in the DMA market advertise robotic surgery, this advertising may have a spillover effect on patients at a focal hospital. Column 3 shows the result with the focal hospital's robotic surgery and nonrobotic surgery advertisements, as well as other hospital robotic advertisements in the DMA market, and we find no evidence of spillover effects.

Heterogeneous advertising effects. To determine whether the advertising effect differs according to patient heterogeneity, we estimate our main model by including all the possible interaction terms between the advertising variable and the patient characteristics variables. The results are reported in Table 4. Patient age, gender, and income show no statistically significant heterogeneity. However, patients classified as Hispanic or other ethnicity are statistically less responsive to robotic surgery advertising than are white patients at 10% and 5% significance levels, respectively. As for insurance status, Medicaid patients are more likely to respond to robotic surgery advertising than Medicare patients. One possible explanation is that robotic surgery may be less familiar to Medicaid patients compared with Medicare patients because of their socioeconomic status. However, once Medicaid patients become aware of its value through advertising, they may be more likely to choose it because their insurance actually covers this procedure.

Table 3. Parameter Estimates for Response to Advertising Content.

Model Specification	(1) Main	(2) Placebo	(3) Spillover
Robot ads (\$/100 capita)	.139*** (.026)		.139*** (.024)
Nonrobot ads (\$/100 capita)		.003 (.004)	.002 (.005)
Market robot ads (\$/100 capita)			.020 (.021)
Market nonrobot ads (\$/100 capita)			.002 (.001)
Number of observations	43,700	43,700	43,700
R ²	.596	.596	.596

* $p < .1$, ** $p < .05$, *** $p < .01$.

Notes: The dependent variable represents a dummy variable that equals one if robotic surgery is chosen, and zero if laparoscopic surgery (or open surgery for prostate) is chosen. Nonrobot ads indicates a focal hospital's advertising expenditures without robotic surgery ad creatives. Market robot ads indicates robotic surgery advertising by other hospitals in the DMA. Market nonrobot ads indicates all other content of hospital advertising by other hospitals in the DMA. All models include patient control variables (demographics, insurance, and comorbidity index), hospital-department-border-county fixed effects, and hospital-department-border-quarterly fixed effects. Complete parameter estimates are included in Web Appendix G. Robust standard errors clustered at the hospital level are reported in parentheses.

Patients with a higher comorbidity index tend to be less responsive to advertising ($p < .1$), potentially because they place more weight on medical circumstances than on advertising or because they simply have more prior knowledge of different types of surgery. However, this effect is not statistically significant at the 5% significance level.

We also observe substantial heterogeneous ad effects across medical departments, with the largest effect being in the ObGyn department. See Web Appendix H for more details.

Patient Health Outcome and Costs Results

Our finding that robotic surgery advertising affects patient choice of robotic surgery raises the question of whether patient outcomes and the cost of robotic surgery differ from those of nonrobotic alternatives. As mentioned previously, we estimate Equation 2 with various empirical specifications to determine if the choice of robotic surgery is associated with different dependent variables of health outcomes and charges (α_1 in Equation 2). Table 5 summarizes the parameter estimates of robotic surgery. First, analyses with LOS (logarithmic function of LOS, to be precise) as a dependent variable show generally negative coefficients for robotic surgery (an approximately 6%–11% reduction at 1% statistical significance) compared with nonrobotic surgeries. For example, the OLS model implies a reduction of 5.9% (95% CI: [1.8%, 10.0%]) in 3.43 (days) average LOS, or 4.86 hours. Specifications using match methods (PSM, CEM), accounting for selection bias based on rich observed patient characteristics and surgery characteristics, indicate greater reductions of approximately 11%.

Table 4. Parameter Estimates for Heterogeneous Advertising Effects.

Ad Variable		Robot Ads (\$/100 Capita)				
Estimate		.186***				
SE		(.070)				
Demographic Variables	Ads × Age	Ads × Female	Ads × Income (\$000)	Ads × Black	Ads × Hispanic	Ads × Other Ethnicity
Estimate	4.34×10^{-5}	-.012	5.90×10^{-5}	.015	-.147*	-.050**
SE	(.001)	(.033)	(.002)	(.041)	(.086)	(.020)
Insurance and Medical Condition Variables	Ads × Medicaid	Ads × Private	Ads × Other Insurance	Ads × Comorbidity Index		
Estimate	.162***	-.031	-.019	-.009*		
SE	(.057)	(.062)	(.044)	(0.005)		

* $p < .1$, ** $p < .05$, *** $p < .01$.

Notes: The dependent variable represents a dummy variable that equals one if robotic surgery is chosen, and zero if laparoscopic surgery (or open surgery for prostate) is chosen. The model includes patient control variables (demographics, insurance, and comorbidity index), hospital-department-border-county fixed effects, and hospital-department-border-quarterly fixed effects. The reference ethnicity is white, and the reference insurance is Medicare. Robust standard errors clustered at the hospital level are reported in parentheses. $N = 43,700$. $R^2 = .596$.

Table 5. Patient Health Outcomes and Cost.

A: LOS, 30-Day Readmission, Bleeding									
DV	LOS			30-Day Readmission			Bleeding		
Model	OLS	PSM	CEM	OLS	PSM	CEM	OLS	PSM	CEM
Robot	-.059*** (.021)	-.110*** (.019)	-.109*** (.019)	.001 (.003)	-.003 (.005)	<.001 (.005)	<-.001 (.005)	-.005 (.006)	-.007 (.006)
N	42,691	22,453	35,764	42,691	22,453	35,764	42,691	22,453	35,764
R ²	.287	.269	.275	.023	.019	.020	.056	.022	.029
B: Mortality Rates, Total Charge									
DV	In-Hospital Mortality			Two-Year Mortality			Total Charge		
Model	OLS	PSM	CEM	OLS	PSM	CEM	OLS	PSM	CEM
Robot	-.001 (.001)	<.001 (.001)	-.001 (.001)	-.003 (.002)	-.005 (.003)	-.003 (.002)	.224*** (.027)	.161*** (.028)	.157*** (.029)
N	42,691	22,453	35,764	42,691	22,453	35,764	42,691	22,453	35,764
R ²	.018	.013	.016	.037	.024	.032	.506	.156	.154

* $p < .1$, ** $p < .05$, *** $p < .01$.

Notes: DV = dependent variable. Parameter estimates for robot are represented by α_1 in Equation 2. For LOS and Total Charge, logarithmic functions of LOS and total charges are used as DVs. All models include patient risk factor controls (demographics, insurance, and medical conditions) and department dummy variables as covariates. The OLS models additionally include hospital and quarterly fixed effects. Robust standard errors clustered at the hospital-level are reported in parentheses. Our original sample size of 43,700 is reduced to 42,691 because some patients could not be tracked for long-term outcomes such as two-year mortality.

Matching details, including postestimation results, are included in Web Appendix I.1. These results are consistent with those in the medical literature (Bhama et al. 2016; Paek et al. 2016). A shorter LOS is often touted in robotic surgery advertising as the major advantage of this surgery, partly due to smaller incisions and less tissue trauma. A back-of-the-envelope calculation suggests that an additional \$1 per 100 capita spent on robotic surgery advertising in a quarter translates to \$25.95 for patients due to a shorter LOS (Web Appendix I.2).

Second, we do not observe a statistically significant relationship between 30-day readmission and robotic surgery at the 5% significance level. This is consistent across three different specifications (OLS, PSM, and CEM). We observe similar results for bleeding, in-hospital mortality, and two-year mortality rates, implying that, except for LOS, we cannot reject the null hypothesis that robotic surgery and nonrobotic surgery health outcomes, including long-term outcomes such as 30-day readmission and two-year mortality, are the same. It is important to note that these adverse health outcomes are rare events.

Therefore, the observation of nonsignificance may be attributed to the lack of statistical power. Despite the lack of statistical significance, we calculate the economic significance of robotic surgery to be \$19,680 due to the two-year mortality reduction (Web Appendix I.2).

We conduct similar analyses with surgery charges. The last three columns of Panel B in Table 5 present the difference between the two types of surgeries in terms of a logarithmic function of total charges. Robotic surgery is 16%–22% more expensive than the nonrobotic alternative, depending on specifications. For example, CEM results imply that the charge for robotic surgery is 15.7% (95% CI: [10.0%, 21.4%]) higher than that for nonrobotic surgery, where the average charge across both surgery types is \$75,683. This result is also in line with that in the medical literature (Jeong et al. 2017; Wright et al. 2013). Despite the cost savings from shorter LOS, robotic surgery is generally considered more costly than laparoscopic surgery due to longer setup time and more expensive consumable equipment costs (Hussain et al. 2014; Kim, Kim, and Baik 2014).

Note that the OLS model includes hospital and quarterly fixed effects, accounting for time-invariant hospital-level factors and quarter-specific factors. Questions may arise regarding hospital-specific time-changing factors (e.g., the composition of physicians and their adoption and learning with robotic surgery may evolve differently over time across hospitals) that could bias our results. We show in Web Appendix I.3 that our results are consistent with the inclusion of hospital-quarterly fixed effects to account for the confounding factors. In the absence of an RCT, our claim regarding these outcomes using matching methods may still be subject to selection bias due to unmatched unobservables. However, conducting an RCT with a large number of patients across various departments is not feasible due to ethical and public health concerns. Nevertheless, the finding of qualitatively and quantitatively consistent shorter LOS and higher costs with robotic surgery across different empirical specifications alleviates concerns that our results may be substantially biased.

Although determining the overall value of robotic surgery in a perfect causal manner beyond matching methods is difficult and outside the scope of this study given our data, these supplementary results suggest that robotic surgery advertising that sways patients to choose robotic surgery may ultimately have implications for patient health outcomes such as LOS and cost.

Conclusion and Implications

We conclude this study by providing a summary of findings and implications for various stakeholders, followed by limitations and directions for future research.

Summary

In the current era of health care consumerism and marketing, the literature has highlighted the effectiveness of advertising in promoting health care consumption in such categories as prescription, insurance, and hospital choice. Interviews with hospital

marketing departments revealed that the return on investment for hospital advertisements is difficult to measure. This is not surprising given recent literature showing that advertising is not always effective in many product categories (Shapiro, Hitsch, and Tuchman 2021), including the case of health insurance ads (Shapiro 2020). Despite remarkable growth in the last two decades, however, the DTCA of hospitals and health care systems has been understudied. To the best of our knowledge, ours is the first empirical study to measure the advertising effectiveness of advertising for high-tech medical procedures based on causal inference.

Using approximately 140,000 patient-level minimally invasive surgery records from hospitals in Florida between 2011 and 2015, we quantify a meaningful, positive impact of robotic surgery advertising on the choice of this procedure as opposed to the alternative surgery types (open surgery for prostatectomy and laparoscopy for other procedures). We find that the average spending of \$.258 per 100 capita on robotic surgery ads per quarter increases the probability that a patient will choose robotic surgery by 3.6% (95% CI: [2.3%, 4.9%]). We conduct supplementary analyses and show that robotic surgery is associated with shorter LOS, but at a higher cost. However, we find no statistical significance for other health outcomes.

Consumer Implications

Critics argue that marketing that promotes high-tech procedures can influence patient preferences, but it does so without providing sufficient information on the pros and cons of the procedure. For instance, a 2019 FDA warning about using robotic surgeries for women undergoing mastectomy and other cancer-related surgeries stated that “patients may not be aware that the safety and effectiveness of these devices have not been established for use in mastectomy procedures or the prevention or treatment of cancer” (FDA 2019). Given the potential impact of robot surgery advertising on patients, a critical need exists for empirical research using individual patient data that can help regulators better understand how such promotions can affect consumer choices. Our results indicate that DTCA is an effective way to sway patient choice, even for decisions involving direct and substantial health risks. To the best of our knowledge, our study is the first to quantify that the advertising of high-tech surgery can influence consumers to choose this type of surgery. For patients, we do not observe long-term health benefits (e.g., 30-day readmission, two-year mortality) associated with robotic surgery. However, the shorter LOS associated with robotic surgery (approximately five hours shorter) is a clear benefit to patients. As long as the patient’s out-of-pocket costs are low (e.g., no or low deductible), robotic surgery advertising can improve consumer welfare.

Managerial Implications

Our study offers managerial implications for several distinctive stakeholders in the health care sector.

Health care providers. We document that patients are influenced to choose robotic surgery as a result of advertising. In addition, the higher charge for robotic surgery benefits health care providers such as hospitals and individual physicians. More importantly, we offer new insights into the development and deployment of effective advertising related to high-tech procedures through heterogeneous advertising effects across patient characteristics; for example, Medicaid patients are more responsive to robotic surgery television advertising. This insight can be used by marketers for providers to better allocate marketing resources and efficiently reach consumers. In addition, patients discern and respond to high-tech ad creatives. Our placebo test shows that the choice of robotic surgery is not affected by hospital advertising that does not focus on this procedure. Therefore, marketers should consider consumer responses to specific advertising content when developing marketing plans.

Surgical robot manufacturers. Surgical robot manufacturers can use our findings to consider investing in cooperative advertising with hospitals as a B2B marketing strategy. Consumer packaged goods brands have long offered cooperative advertising budgets to retail partners, along with advertising content requirements and restrictions. Some manufacturers have engaged in “ingredient branding” in B2B settings to build brand equity among consumers (e.g., Intel advertising directly to consumers). Therefore, investment in producing their own robotic surgery videos and claims can help surgical robot manufacturers expand the market and build brand recognition and affinity.

Insurance providers (payers). The implications for payers such as private insurance companies and state and federal governments depend on the net effect of patient short-term and long-term health outcomes and costs. Robotic surgery ads can hurt insurance providers that offer plans with low out-of-pocket costs (e.g., little or no deductible), especially when the robotic surgery’s shorter LOS is deemed minimal. Our results also imply that robotic surgery advertisements have a greater influence on Medicaid patients who tend to be of lower socioeconomic status. This may be of interest to the federal and state governments funding Medicaid, as this group tends to lack resources to manage their health. More information disclosure on risks and alternatives in advertising, similar to that required for pharmaceutical drug DTCA, could potentially educate viewers and offer some level of protection for the disadvantaged consumer segment in this market.

Policy Implications

Hospital advertising in the United States has a controversial history. Until 1980, hospital advertising was banned by the American Medical Association, which considered it to be “derogatory to the dignity of the profession” (Tomycz 2006, p. 26). However, in 1980 a circuit appellate court ruled that this ban violated the Federal Trade Commission Act protecting free commerce. While hospital advertising is legal now, it has been the

focus of ongoing debate. Our findings have important implications for health care policy makers because patients are affected by robotic surgery advertising, but no clear evidence in our study or other medical studies demonstrates the superiority of robotic surgery regarding long-term treatment outcomes (Jayne et al. 2017; Jeong et al. 2017). By examining the role of advertising and consumer response, our finding of a causal effect of advertising on the choice of robotic surgery provides new insight into the debate regarding the value of robotic surgery.

It is important to note that deceptive, false claims in hospital advertising are subject to investigation and prosecution under the Federal Trade Commission Act.³ However, unlike DTCA of prescription drugs, DTCA of robotic surgery is not overseen by the FDA. Although the FDA regulations require drug manufacturers to offer a balanced view of the risks and benefits in advertisements, DTCA of robotic surgery lies outside the scope of this regulation. As a consequence, television DTCA for medical procedures or medical devices requires no disclosure of either benefits or risks in the typical 30-second advertisement format.

In light of our results that robotic surgery advertising can materially increase demand for the procedure, along with the complexity of surgical operations and their risk for patients, regulators might consider imposing rules for the marketing of high-tech procedures that are similar to those for pharmaceutical DTCA, which require disclosures of the pros and cons of the procedures, including possible surgery complications. This would allow patients to make more informed decisions about the choice of surgery type. In fact, Cancer Treatment Centers of America voluntarily includes disclaimers in its advertisements. Its advertisements also encourage viewers to visit the company website so that they can examine clinical data on treatment results for 11 types of cancer (*Knowledge at Wharton* Staff 2014). Given the risks associated with surgery, our findings on advertising effectiveness point to the need for further policy discussion on the transparency of advertising content.

Limitations and Future Research

This study has several limitations. First, our data are limited to Florida hospital patients. Even though Florida is the third most populous state in the United States, its population is substantially older than the average (sixth oldest in the United States). Therefore, generalizing our results to the rest of the country is difficult. Future studies could acquire inpatient data from other states and apply our approaches to confirm our findings.

Although we examine the content of advertisements by checking whether they promote robotic surgery, we do not capture other aspects of their content. For instance, would consumers respond differently if the advertisements contained different ad copies and messages? What content in hospital advertising is the most effective? Future research can further

³ Rao and Wang (2017) show that the revelation of false claims due to settlements with the FTC decreases demand in four grocery product categories.

study the content of advertising focused on high-tech procedures to help markets engage in more effective promotions and to inform regulators. A final limitation is that we mainly focus on the patient response to advertising without delving into the effect of hospital competition and patient welfare. Future research could shed light on the effectiveness of advertising in relation to these two issues and robotic surgery choices.

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